

Cycle financing and a period lower bound for the Juggler map

Philippe Cochin
philippe@cochin.fr

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

It is conjectured that every positive integer eventually reaches 1.

We develop a cycle-financing inequality for this floor-power map. Exact integer cells give a one-step logarithmic defect; cycle minimality lets that defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. For a hypothetical cycle of length L with o odd steps and minimum n ,

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

As a consequence, combining the inequality with the known verification through 10^6 yields $L \geq 25781$. The novelty is not a new computational record; it is that implication. We also prove that every nontrivial cycle contains at least four even steps, and hence has period at least eleven. The core arguments are formalized in Lean 4; the finite numerical tables are independently certified computations.

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1. Introduction

The Juggler sequence was introduced by Pickover [1,2]. The one-step map is OEIS A094683 [3]; the stopping-time table is A007320 [4]. The orbit of 3 is 3, 5, 11, 36, 6, 2, 1. The orbit of 37 already peaks at 24906114455136. Universal arrival at 1 remains the open conjecture of the subject.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \quad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities.

The mechanism is the interaction of three elementary facts: exact integer cells, a logarithmic defect, and cycle minimality. Exact cells give a one-step logarithmic defect; cycle minimality lets the defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. That is Theorem 4.4:

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

The inequality $\log(1+u) \leq u$ is the only analytic input; it is not the content of the theorem.

Write $N_0 = 10^6$ for the *verified descent floor* reported by Weisstein [5] and recomputed here by exact first-passage. That floor is an input, not the result. Combined with Theorem 4.4 it gives

$$\text{known verification through } 10^6 + \text{this inequality} \Rightarrow L \geq 25781.$$

The architecture is

$$\text{envelope} \rightarrow \text{cycle minimum} \rightarrow \text{finance} \rightarrow n_{\max}(L) \rightarrow N_0 = 10^6 \rightarrow L \geq 25781.$$

The computation supplies the endpoint 10^6 . The mathematics amplifies that floor to the period bound.

Roadmap. Section 2 records the power envelope and the exact defect identity that explains it. Section 3 classifies minimum-based cycle words and proves that every nontrivial cycle has at least four even letters. Section 4 unrolls the cell logarithm around a cycle minimum, obtains the finance inequality, and applies it at the known floor 10^6 . A run-type packing of the same identity is a supporting refinement; the arithmetic of the leftover lengths is secondary.

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$. Write J^k for the k -fold iterate. A nonempty realized word w with $J^{|w|}(n) = n$ is a *cycle word*. The unique fixed point is 1; a cycle is *nontrivial* when it contains some $n \geq 2$. A cycle word at n is *minimum-based* when n is a cycle minimum: $J^j(n) \geq n$ for every $0 \leq j < |w|$. Every cycle word has a minimum-based rotation.

All exceptional sets below are *finance-survivor* sets: lengths that a stated form of the finance inequality does not exclude at the verified descent floor. They are not candidate cycle sets.

1.0 Main results

Contribution 1 — cycle-financing inequality. For a minimum-based cycle of length L with o odd steps,

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o$$

(Theorem 4.4). Floor defects become a quantitative bound on the cycle minimum.

Contribution 2 — structural word obstruction. Every nontrivial cycle word has at least four even letters, and hence period at least eleven (Theorem 3.22 and Corollary 3.23). The argument classifies the minimum-based word geometry; it is not a raw census of words of length at most ten. This bound does not use the verified descent floor.

Contribution 3 — explicit global consequence. Combined with the independently verified descent floor $N_0 = 10^6$, there is no nontrivial Juggler cycle of length at most 25780. Equivalently, any nontrivial cycle has period at least 25781 (Theorem 4.6(A)).

These three statements are not interchangeable. Theorem 4.4 is the conceptual sharp inequality (constant 1). Corollary 4.5 is the convenient length-only statewise bound that turns a descent floor into a per-length exclusion. Theorem 4.6 is the numerical certification of that bound, and it uses a conservative coefficient $6/5$ only as a uniform majorant; the headline cutoff 25781 is not an artifact of that majorant. Theorem 4.7 is a run-type refinement of the same defect sum. The leftover lengths through 10^5 are supporting material, not a second main theorem.

1.1 Related work

Pickover's later exposition is Chapter 45 of [2]. Weisstein [5] records the map, the stopping-time sequences A094670, A094679, A095908, and a verification of arrival at 1 through 10^6 . That verified descent floor is the computational input to Theorem 4.6; the first-passage run of Appendix B recomputes it. OEIS A094716 [6] records extreme heights, including the start 48443 whose peak has 972 463 digits. Those height records do not bound the period.

Prasad–Prasad [7] estimate excursion and stopping constants for juggler-like maps by a random-walk large-deviation model; those estimates do not apply to exact cycles. Small-cycle censuses are a standard first layer for Collatz-like maps, surveyed by Lagarias [8,9]. Those results do not transfer: the branches of J are floor powers rather than affine maps (Crandall [10], Matthews–Watts [11]). In particular there is no identity of the form $n(2^K - 3^p) = C$. A check of Pickover [1,2], Weisstein [5], the OEIS records [3,4,6], the Prasad–Prasad estimates [7], and the standard Collatz cycle-bound sources [8–13] found no published explicit lower bound

on the period of a nontrivial cycle for this exact floor-power Juggler map. Informal near-miss notes and later unverified webpage floors are not used.

The layers of the argument are as follows.

1. *Classical idea*: cycle financing (Simons–de Weger [12]).
2. *Classical idea*: logarithmic and continued-fraction approximation of $\log 2 / \log 3$.
3. *Classical idea*: cycle-word restrictions and leftover packaging (Eliahou [13]; Lagarias [8,9]).
4. *New object*: the exact one-step floor-power cells of J .
5. *New theorem*: the Juggler-specific cycle-minimum finance inequality of Theorem 4.4.
6. *New consequence*: $L \geq 25781$ at the verified descent floor 10^6 .
7. *Supporting organization*: the run-packing refinement of the same defect sum (Theorem 4.7).

Financing as an idea, logarithmic step inequalities, and the continued-fraction phenomenon are not claimed as new. The argument below is elementary and independent of the Diophantine tools of [12]: floor-power defects are relatively $O(1/x)$ in logarithms, so a uniform logarithmic floor-error bound, valid above the verified floor, excludes every length that is not a finance-survivor for Theorem 4.4. To the best of our knowledge, the Juggler-specific inequality and the explicit period bound it produces are new.

Novelty statement. For this nonlinear floor-power map, exact floor defects convert into a cycle-financing inequality that forces the minimum of a hypothetical cycle below the independently verified descent region unless the period is at least 25781.

1.2 Verification

The arguments of Sections 2, 3, and 4 may be read without machine assistance. The core mathematical lemmas are mechanized in Lean 4; selected finite classifications and numerical tables are independently certified computations. Appendix A records the Lean names. The finite tables used by Lemmas 3.5, 3.7, 3.11 and Theorems 3.12–3.20 are `native_decide` evaluations in the modules named there.

Proposition 1.3 (certified computational input). A machine-verifiable certificate establishes that every integer $2 \leq n \leq 10^6$ reaches 1. Precisely: an exact-integer first-passage run records, for each such n , a finite realized word with image strictly below the start; strong induction on that image reaches 1. The longest first passage in the window has 253 steps (seed 78901). Weisstein [5] records the same computational verification; the run here is an independent recomputation. The certificate files, SHA-256 hashes, and regeneration commands are Appendix B.

Roles. Independently proved: the finance inequality (Theorem 4.4). Computational input: every $2 \leq n \leq 10^6$ reaches 1 (this proposition). Independently recomputed: the exact first-passage run of Appendix B. Not proved: global termination.

Theorem 4.6 applies Corollary 4.5 to this input and certifies the table with the conservative coefficient $6/5$. Theorem 4.7 is a human proof; Theorem 4.8 reuses the gap table under that packing. Proposition 4.9 is integer arithmetic in Lean. This is not a claim that the paper as a whole is formally verified.

```
Repository: https://github.com/sneakyweasel/balanced_ternary
Commit:    7802f78bec58c68cec92a2efb3db4a502f916277
Lean:      leanprover/lean4:v4.33.0
Mathlib:   v4.33.0 (lake-manifest rev db584cd6d46c92f209a44c0f1c829460d327499d)
Build:     lake build Problems.JugglerPaper (from formal/)
Computation: python -m research.juggler_sequence.cycle_finance
              (parity table); run-type table in budget_opt.json
SHA-256:   Appendix B
```

The commit is the repository state that produced the finance tables. A later editorial commit of this text does not change those hashes.

2. Envelope

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w . Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters. When w is realized, this endpoint is the $|w|$ -fold iterate.

Theorem 2.1 (fixed-word monotonicity). If $n \leq m$ and both realize w , then $J^{|w|}(n) \leq J^{|w|}(m)$.

Proof. Induct on w . The empty word is immediate. The images after a common realized prefix remain ordered, and both current states realize the same next letter. The even branch is the monotone integer square root; the odd branch is $x \mapsto x^3$ followed by the integer square root. $\square$

The realizing set of a fixed word need not be an interval: OE is realized at 7 and at 11 but not at 9.

Theorem 2.2 (finite-word power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

Proof. The empty word is the equality $n \leq n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \leq n^{3^o}$, and the next letter is realized.

If the next letter is even, then $J(x)^2 \leq x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{2^\ell} \leq n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x)^2 \leq x^3$, so

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{3 \cdot 2^\ell} = (x^{2^\ell})^3 \leq (n^{3^o})^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. $\square$

Corollary 2.3 (exponent-gap contraction). If $n \geq 2$, w is realized at n , and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \leq n^{3^{\#O(w)}}$. The exponent gap and $n \geq 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \geq 1$, one has $m < n$. $\square$

The corollary includes familiar contracting blocks such as $OOOEE$. It does not prove that every start realizes some contracting word.

2.4 Exact floor defect

Theorem 2.2 is the inequality form of an exact identity

$$n^{3^{\#O(w)}} = J^{|w|}(n)^{2^{|w|}} + \Delta_w(n), \quad \Delta_w(n) \geq 0.$$

The identity, its vanishing law, and the two-term composition are Theorems 2.4–2.6 and Corollary 2.7 in Appendix C. A mixed realized word has $\Delta_w(n) > 0$. This explains why the envelope of Theorem 2.2 is true, and why a mixed cycle word is formally expanding. Theorem 4.4 uses only the nonnegativity $\Delta_w(n) \geq 0$.

No uniform local tax exists: the relative slack of a single letter tends to 0 with the state. Finance must therefore be a global comparison, not a fixed cost per odd or even step. A quantitative itinerary-dependent lower bound on Δ_w , or a tighter upper bound on $\sum 1/(x_i \log x_i)$, would improve the period cutoff; neither is proved here.

3. Structural restrictions on cycle words

The headline of this section is Theorem 3.22: every nontrivial cycle word has at least four even letters, and hence period at least eleven. The small-period censuses (Theorems 3.6 and 3.8) are supporting. After the one-step cells, a minimum-based word has a canonical run form. There are only three even-count regimes with $e \leq 3$; each reduces to a finite collection of geometries $O^aEO^bEO^cE$. Those geometries are eliminated by a next-square obstruction, by long odd-run growth against a last-even cell, or by a finite exceptional window. The family calculations are Appendix D.

The exact one-step fibers are

$$J(n) = q \iff q^2 \leq n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \leq n^3 < (m+1)^2 \quad (n \text{ odd}).$$

Lemma 3.1 (odd cells are unique). An odd fiber contains at most one integer. An even fiber is a parity-restricted square interval and may contain many predecessors.

Proof. Suppose $a < b$ lie in the same odd cell indexed by m . Then $(a+1)^3 \leq b^3 < (m+1)^2$ and $m^2 \leq a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

hence $3a^2 + 3a < 2m$ and $3a^2 < 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \leq a^3$ gives $9a^4 < 4m^2 \leq 4a^3$, contradicting $4a^3 < 9a^4$. $\square$

Theorem 3.2 (cycle restrictions). Let w be a cycle word at $n \geq 2$.

- (i) The word is formally expanding:

$$2^{|w|} < 3^{\#O(w)}.$$

A contracting word cannot close a nontrivial cycle.

- (ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter.

- (iii) A realized word v is *superquadratic* if

$$3^{\#O(v)} \geq 2^{|v|+1}.$$

Any realized path from a start $n \geq 2$ to a state at least n^2 is superquadratic. On a cycle minimum the path to any later even state is superquadratic. The prefix OOE is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \leq n^{3^{\#O(w)}}$. Since $n \geq 2$, comparison of exponents gives $2^{|w|} \leq 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \geq 1$. Alternatively: a mixed cycle word has $\Delta_w(n) > 0$ by Theorem 2.5 in Appendix C, so the envelope is strict; a monochrome tower cannot return for $n \geq 2$.

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start $n \geq 2$. An even state $x \geq 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \geq 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n . If the last letter were odd, the odd return cell would give $n^2 \leq x^3 < (n+1)^2$. Minimality gives $x \geq n$, hence $n^3 < (n+1)^2$, impossible for $n \geq 3$; and the minimum is odd, so $n \geq 3$.

For (iii), let a realized word v send n to $y \geq n^2$. Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since $n \geq 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n . Thus $n \leq J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \leq y$, and the preceding argument applies to that prefix. $\square$

Lemma 3.21b (canonical run form). After rotation to a minimum-based orientation, the word begins with OO and ends with E ; hence every word with $1 \leq e \leq 3$ even letters has the canonical run decomposition $O^a EO^b EO^c E$ (unused runs empty). The case $e = 0$ is all-odd and is already forbidden by the last-letter restriction. No other cyclic rotation needs a separate case.

Proof. Theorem 3.2: the minimum is odd, so the word cannot begin with E or OE , and it cannot end with an odd letter. Thus a minimum-based word starts OO and ends E , so $e \geq 1$. The remaining letters are odd runs separated by the e even letters, so the word is $O^{a_1} E \dots O^{a_e} E$ with $a_1 \geq 2$. For $e \leq 3$ this is $O^a EO^b EO^c E$ with unused runs empty. Every cycle word has a minimum-based rotation of the same even-count, and that orientation is already in this form. $\square$

Thus it is enough to exclude the three regimes $e = 1, 2, 3$. The rest of the section does that. Theorems 3.6 and 3.8 record the short-period consequences; Theorem 3.22 is the structural statement.

Lemma 3.3 (coarse lower envelope). For $n \geq 1$, write $q = \lfloor \sqrt{n} \rfloor$. Then $q^2 \leq n < (q+1)^2$. For $q \geq 1$ one has $(q+1)^2 \leq 4q^2$, and the second inequality is strict for $q \geq 2$, while for $q = 1$ one has $n < 4$. In all cases

$$n < 4 \lfloor \sqrt{n} \rfloor^2.$$

Thus an even step satisfies $n \leq 4J(n)^2$ and an odd step satisfies $n^3 \leq 4J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o} \leq C_v J^k(n)^{2^k}.$$

The constant starts at 1 and updates by $C \mapsto C \cdot 4^{2^j}$ on an even letter and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j . In particular $C_{OOO} = 2^{38}$ and $C_{OOOO} = 2^{130}$.

Proof. The comparison $n < 4q^2$ is the paragraph above. The composition law is the same recurrence as the proof of Theorem 2.2, with a factor 4^{2^j} inserted at each letter. The values C_{OOO} and C_{OOOO} are the result of that recurrence on those two words. $\square$

Lemma 3.4 (next-square thresholds). (i) If $q \geq 5$ realizes OO , then $J^2(q) \geq (q+1)^2$. (ii) If $q \geq 3$ realizes OOO , then $J^3(q) \geq (q+1)^2$. (iii) If a realized word v satisfies $J^{|v|}(q) \geq (q+1)^2$ and the next realized letter is odd, then $J^{|v|+1}(q) \geq (q+1)^2$. (iv) If vE is a cycle word at n , then $J^{|v|}(n) < (n+1)^2$. (v) If $a \geq 3$, then $O^a E$ is not a cycle word at any $n \geq 2$.

Proof. For (i), write $m = \lfloor q^{3/2} \rfloor$. The bound $J^2(q) \geq (q+1)^2$ follows from $m^3 \geq (q+1)^4$, because then $\lfloor m^{3/2} \rfloor \geq (q+1)^2$. If $q = 5$, then $11^2 = 121 \leq 125 = 5^3 < 144 = 12^2$, so $m = 11$, and $11^3 = 1331 \geq 6^4 = 1296$. Since q realizes OO , it is odd, so the only remaining case is $q \geq 7$. Then $m \geq q^{3/2} - 1$, so it is enough that $(q^{3/2} - 1)^3 \geq (q+1)^4$. Expanding the left side and dropping the positive remainder $3q^{3/2} - 1$ reduces this to $q^{9/2} - 3q^3 \geq (q+1)^4$, or equivalently $\sqrt{q} - 3/q \geq (1 + 1/q)^4$. The left side increases for $q > 0$ and the right side decreases, so the case $q = 7$ suffices:

$$\sqrt{7} - \frac{3}{7} > \frac{5}{2} - \frac{3}{7} = \frac{29}{14} > \frac{4096}{2401} = \left(\frac{8}{7}\right)^4,$$

where $\sqrt{7} > 5/2$ follows from $25/4 < 7$.

For (ii), the orbit $3 \rightarrow 5 \rightarrow 11 \rightarrow 36$ gives $J^3(3) = 36 \geq 16$. If $q \geq 5$ realizes OOO , then it realizes OO , so (i) gives $J^2(q) \geq (q+1)^2$. The third letter is odd, hence $J^3(q) = J(J^2(q)) \geq J^2(q)$.

For (iii), the image after v is odd and at least $(q+1)^2 \geq 4$, so the odd branch does not decrease it.

For (iv), the last letter is even, so the preimage $z = J^{|v|}(n)$ is even and satisfies $n^2 \leq z < (n+1)^2$.

For (v), suppose $O^a E$ is a cycle word at $n \geq 2$. The start is odd, hence at least 3, and realizes O^a . Parts (ii) and (iii) give $J^a(n) \geq (n+1)^2$, contradicting (iv). $\square$

Lemma 3.5 (two length-six exclusions). Neither $OOOEOE$ nor $OOOOEE$ is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2 \cdot 256^{64}$ because $(1+1/256)^{64} < e^{64/256} = e^{1/4} < 2$, and $256(n+1) \leq 257n$, so $(n+1)^{64} < 2n^{64}$. Then $2^{130}(n+1)^{64} < 2^{131}n^{64}$. Since $n \geq 256 = 2^8$, one has $n^{17} \geq 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \leq n^{81}$.

For $2 \leq n < 256$, neither word returns to its start. This is a table of 254 evaluations of each word: at every start, either some letter fails to match the current parity, or the six-step image differs from the start. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooee` and `no_cycle_word_ooooee` (Appendix A).

Now suppose $OOOOEE$ is a cycle word at $n \geq 256$, and write $z = J^4(n)$ for the image after the prefix $OOOO$. Lemma 3.4(iv) gives $J(z) < (n+1)^2$, and the preceding even letter gives $z < (J(z)+1)^2$, hence $z < (n+1)^4$. Lemma 3.3 on $OOOO$ gives $n^{81} \leq 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $OOOEOE$ is a cycle word at $n \geq 256$. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor \sqrt{z_3} \rfloor$, so $z_3 < (y+1)^2$. Lemma 3.3 on OOO gives $n^{27} \leq 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 257$. We claim $(y+1)^3 < 2A^4$. If $y \leq A$, this is $(A+1)^3 < 2A^4$. If $y > A$, then $y \geq 258$, so $3y+1 < y^2$ and hence $(y+1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \leq yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. $\square$

Theorem 3.6 (small-cycle census). No word of length at most six is a cycle word at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word; every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start $n \geq 2$.

If every letter is odd, the start is odd, hence $n \geq 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \geq 3$, since $x^3 \geq (x+1)^2$. The orbit ascends strictly and never returns. Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word vE of the same length based at a cycle state $m \geq 2$. It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.2(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ($2 > 3^0$ and $4 > 3$).

Length three: the only expanding candidate is OOE . If $m \geq 5$ realizes OO , Lemma 3.4(i) gives $J^2(m) \geq (m+1)^2$, contradicting Lemma 3.4(iv). The only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd, so the final even letter is not realized.

Length four: the expanding filter requires three odd letters among the first three, leaving only O^3E , excluded by Lemma 3.4(v).

Length five: the filter requires four odd letters among the first four, leaving only O^4E , excluded by Lemma 3.4(v).

Length six: the filter requires at least four odd letters among the first five, leaving O^5E , $EOOOOE$, $OEOOOE$, $OOOEOE$, and $OOOOEE$. Lemma 3.4(v) excludes O^5E . The word $EOOOOE$ rotates one step onto $OOOOEE$, and $OEOOOE$ rotates two steps onto $OOOEOE$; both are excluded by Lemma 3.5.

It remains to exclude $OOEEOE$. Let this word be a cycle word at some start, and rotate to a cycle minimum $m \geq 2$. The three alignments of the two even letters are $OOEEOE$, $OEOOEO$, and $EOOEEO$. The last starts even, so it cannot be a minimum, by Theorem 3.2(ii). The middle starts OE : the first image is even and strictly below m^2 , whereas every even state after a cycle minimum is at least m^2 , as proved in Theorem 3.2(iii). Thus the minimum orientation is $OOEEOE$. In particular m is odd, so $m \geq 3$. The prefix OOE

must be realized. If $m = 3$, then $3 \rightarrow 5 \rightarrow 11$ and 11 is odd, so OOE is not realized. Hence $m \geq 5$. Write $y = J^3(m)$ for the state after OOE . Then $y \geq m$ by minimality, so $y \geq 5$. The suffix OO is realized at y , and Lemma 3.4(i) gives $J^2(y) \geq (y+1)^2 \geq (m+1)^2$. The last letter is even, so Lemma 3.4(iv) gives $J^2(y) < (m+1)^2$. $\square$

Lemma 3.4(v) excludes every odd-run-then-even word $O^a E$ with $a \geq 3$, of any length.

Lemma 3.7 (two length-seven exclusions). Neither $OOOOEOE$ nor $OOOOOEE$ is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 14$, then

$$n^{243} > 2^{422}(n+1)^{128}.$$

Indeed $14(n+1) \leq 15n$, so $(n+1)^{128} \leq (15/14)^{128}n^{128}$. The finite comparison $2^{422}15^{128} < 14^{243}$ then yields $2^{422}(n+1)^{128} < 14^{115}n^{128} \leq n^{243}$.

For $2 \leq n < 14$, neither word is realized: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooeoe` and `no_cycle_word_oooooee` (Appendix A).

Now suppose $OOOOOEE$ is a cycle word at $n \geq 14$, and write $z = J^5(n)$ for the image after the prefix $OOOOO$. Lemma 3.4(iv) on the last even letter, together with the preceding even letter, gives $z < (n+1)^4$. Lemma 3.3 on $OOOOO$ gives $n^{243} \leq 2^{422}z^{32}$, so $n^{243} < 2^{422}(n+1)^{128}$, contradicting the tail inequality.

Finally suppose $OOOOEOE$ is a cycle word at $n \geq 14$. Write $z_4 = J^4(n)$ and $y = J(z_4) = \lfloor \sqrt{z_4} \rfloor$, so $z_4 < (y+1)^2$. Lemma 3.3 on $OOOO$ gives $n^{81} \leq 2^{130}z_4^{16} < 2^{130}(y+1)^{32}$. Cubing yields $n^{243} < 2^{390}(y+1)^{96}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 15$. The same comparison $(y+1)^3 < 2A^4$ as in Lemma 3.5 holds at this smaller scale. Raising to the thirty-second power gives $(y+1)^{96} < 2^{32}(n+1)^{128}$. Combining with the cubed lower envelope produces again $n^{243} < 2^{422}(n+1)^{128}$. $\square$

Theorem 3.8 (small-cycle census through length seven). No word of length at most seven is a cycle word at any $n \geq 2$. Equivalently, a nontrivial Juggler cycle, if one exists, has period at least eight.

Proof. The reduction of Theorem 3.6 applies at every length: an all-odd word cannot return, and every mixed cycle word rotates to an even-terminating cycle word based at a cycle state $m \geq 2$. Lengths at most six are Theorem 3.6. It remains to exclude even-terminating cycle words of length seven.

By Theorem 3.2(i) a cycle word is formally expanding. The even-terminating expanding length-seven words are exactly O^6E , EO^5E , OEO^4E , $OOEO^3E$, O^3EO^2E , O^4EOE , and O^5EE . Lemma 3.4(v) excludes O^6E . The word EO^5E rotates one step onto O^5EE , and OEO^4E starts OE , so it cannot be a cycle minimum (Theorem 3.2(iii)) and rotates two steps onto O^4EOE ; both leftovers are excluded by Lemma 3.7.

It remains to exclude $OOEO^3E$ and O^3EO^2E . Rotate either word to a cycle minimum $m \geq 2$. For $OOEO^3E$ the minimum orientation retains the internal even letter followed by the suffix OOO . Then $m \geq 3$, the prefix through that even letter is realized, and Lemma 3.4(ii) at threshold 3 contradicts the last even cell. For O^3EO^2E the same bootstrap applies with suffix OO and threshold 5, once $m = 3$ is removed: at $m = 3$ the state after OOO is even, so the next even letter is not realized. $\square$

Theorems 3.6 and 3.8 are supporting period statements. The structural claim is the even-count exclusion of Theorem 3.22. The same cells organise leftover words by even-count. Write

$$e_a = 2(3^a - 2^a)$$

for $a \geq 0$.

Lemma 3.9 (trailing even run). If a cycle word based at n ends with $r \geq 1$ even letters, the state immediately before that run is strictly less than $(n+1)^{2^r}$.

Proof. The case $r = 1$ is Lemma 3.4(iv). Suppose the claim holds for some $r \geq 1$, and let vE^{r+1} be a cycle word at n . Write $z = J^{|v|}(n)$. The inductive hypothesis applied to the suffix E^r after the first of those even letters gives $J(z) < (n+1)^{2^r}$. The state z is even, so $z < (J(z)+1)^2 \leq ((n+1)^{2^r})^2 = (n+1)^{2^{r+1}}$. $\square$

Lemma 3.10 (odd-run lower envelope). If $n \geq 1$ realizes O^a , then

$$n^{3^a} \leq 2^{e_a} J^a(n)^{2^a}.$$

Proof. Lemma 3.3 supplies a multiplicative constant C_v along any realized word, with $C_\varepsilon = 1$, $C \mapsto C \cdot 4^{2^j}$ on an even letter, and $C \mapsto C^3 \cdot 4^{2^j}$ on an odd letter, at step j . On the pure odd word O^a every letter is odd, so $C_{O^{a+1}} = C_{O^a}^3 \cdot 4^{2^a}$. Writing $C_{O^a} = 2^{e_a}$ with $e_0 = 0$ yields the recurrence $e_{a+1} = 3e_a + 2^{a+1}$, because $4^{2^a} = 2^{2^{a+1}}$. The closed form $e_a = 2(3^a - 2^a)$ satisfies the recurrence and the initial value. $\square$

Lemma 3.11 (seven-odd window). No integer n with $2 \leq n < 256$ realizes the word O^7 .

Proof. This is a table of 254 seven-step evaluations: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_follows_seven_odds_of_1t256` (Appendix A). $\square$

Theorem 3.12 (two-even leftover families). Let $k \geq 6$ and $n \geq 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle word at n .

Proof. Appendix D.

A cycle word w at n is *minimum-based* when n is a cycle minimum: $J^j(n) \geq n$ for every $0 \leq j < |w|$. The remainder after a proper prefix of a cycle word need not itself be a cycle word at the prefix endpoint. The next statement therefore transports the tail inequality of Theorem 3.12, not the cycle-word exclusion at a later start.

Theorem 3.13 (first-even transport). Let $n \geq 2$. No minimum-based cycle word at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. Appendix D.

The hypothesis that the start is a cycle minimum is essential. If $y < n$, the leftover cell is measured against a larger start and need not contradict the tail at y . In particular, Theorem 3.13 does not assert that those words fail to be cycle words at a non-minimum start. That upgrade is Theorem 3.21.

Theorem 3.14 (three trailing evens). Let $a \geq 6$ and $n \geq 2$. The word O^aEEE is not a cycle word at n .

Proof. Appendix D.

Theorem 3.15 (mixed bunched family $EOEE$). Let $a \geq 5$ and $n \geq 2$. The word O^aEOEE is not a cycle word at n .

Proof. Appendix D.

Theorem 3.16 (mixed bunched family $EOOEE$). Let $a \geq 4$ and $n \geq 2$. The word O^aEOOEE is not a cycle word at n .

Proof. Appendix D.

Theorem 3.17 (mixed bunched family $EOOOEE$). Let $a \geq 3$ and $n \geq 2$. The word $O^aEOOOEE$ is not a cycle word at n .

Proof. Appendix D.

Theorem 3.18 (mixed bunched family $EEOE$). Let $a \geq 5$ and $n \geq 2$. The word O^aEEOE is not a cycle word at n .

Proof. Appendix D.

Theorem 3.19 (mixed bunched family $EOEOE$). Let $a \geq 4$ and $n \geq 2$. The word O^aEOEOE is not a cycle word at n .

Proof. Appendix D.

Theorem 3.20 (mixed bunched family $EOOEOE$). Let $a \geq 3$ and $n \geq 2$. The word $O^a EOOEOE$ is not a cycle word at n .

Proof. Appendix D.

Theorem 3.21 (gapped leftovers as cycle words). Let $n \geq 2$. No cycle word at n has the form $O^a EO^b EE$ with $a \geq 2$ and $b \geq 4$, or the form $O^a EO^b EOE$ with $a \geq 2$ and $b \geq 3$.

Proof. Appendix D.

Theorems 3.12–3.21 assemble into an even-count exclusion: no cycle word has fewer than four even letters, so a nontrivial cycle has period at least eleven (Theorem 3.22). Section 4 excludes later periods by financing.

Every integer $2 \leq n < 12$ reaches 1, so a cycle minimum is at least 12. Write e for the number of even letters. Lemma 3.21b is the canonical run form.

Lemma 3.21a (classification). Every minimum-based cycle word with at most three even letters belongs to one of the families excluded by Lemma 3.4(v) and Theorems 3.12–3.21.

Proof. Lemma 3.21b puts the word in the form $O^a EO^b EO^c E$ with $a \geq 2$ and unused runs empty. If $e = 0$, the word is all-odd. If $e = 1$, it is $O^a E$. If $e = 2$, a last run $c \geq 2$ is the internal-even bootstrap of Lemma 3.4, and the remaining shapes are the two-even families of Theorem 3.12. If $e = 3$, again $c \geq 2$ is the bootstrap, while $c \in \{0, 1\}$ is either a gapped leftover (Theorem 3.21) or one of the seven bunched families (Theorems 3.14–3.20). $\square$

	remaining minimum-based	
e	forms	elimination
0	all-odd	cannot return (Theorem 3.6)
1	$O^a E$	next-square; Lemma 3.4(v)
2	$O^a EE$, $O^a EOE$, last run ≥ 2	Theorem 3.12; Lemma 3.4
3	bunched families and gapped leftovers	Theorems 3.14–3.21
≥ 4	not excluded by even-count	finance (Section 4)

Theorem 3.22 (even-count). No word with fewer than four even letters is a cycle word at any $n \geq 2$. Equivalently, a nontrivial cycle word has at least four even letters.

Proof. Every cycle word has a minimum-based rotation, with the same even-count. Lemma 3.21b puts that orientation in canonical run form; Lemma 3.21a names the families. In detail:

If $e = 0$, the word is all-odd and cannot return, as in Theorem 3.6.

If $e = 1$, the word is $O^a E$ with $a \geq 2$. The case $a = 2$ is OOE , excluded in Theorem 3.6. The cases $a \geq 3$ are Lemma 3.4(v).

If $e = 2$, the word is $O^a EO^c E$ with $a \geq 2$. A last odd-run $c \geq 2$ is an internal even letter followed by OO or OOO . Lemma 3.4 at the cycle minimum ($n \geq 12$) contradicts the last-even cell. The remaining shapes are $O^a EE$ and $O^a EOE$. Expansion forces $a \geq 4$ and $a \geq 3$ respectively, so both are Theorem 3.12.

If $e = 3$, the word is $O^a EO^b EO^c E$ with $a \geq 2$. Again $c \geq 2$ is the internal-even bootstrap. The remaining last runs are $c = 0$ and $c = 1$. For $c = 0$ and $b \geq 4$ the word is a gapped leftover $O^a EO^b EE$, excluded by Theorem 3.21. For $c = 0$ and $b \leq 3$ the word is one of $O^a EEE$, $O^a EOEE$, $O^a EOOEE$, $O^a EOOOEE$, excluded by Theorems 3.14–3.17 once expansion supplies the stated lower bounds on a . For $c = 1$ and $b \geq 3$ the word is a gapped leftover $O^a EO^b EOE$, excluded by Theorem 3.21. For $c = 1$ and $b \leq 2$ the word is one of $O^a EEOE$, $O^a EEOEOE$, $O^a EOOEOE$, excluded by Theorems 3.18–3.20.

Thus $e \leq 3$ is impossible. $\square$

Corollary 3.23. A nontrivial cycle, if one exists, has period at least eleven.

Proof. Theorem 3.22 gives four even letters, hence $o \leq L - 4$. Formal expansion is $2^L < 3^o \leq 3^{L-4}$. The comparison $2^L < 3^{L-4}$ first holds at $L = 11$. $\square$

In particular there is no cycle of length eight, nine, or ten.

4. Cycle finance

A cycle word is formally expanding (Theorem 3.2), yet the orbit returns exactly. The multiplicative surplus $3^o - 2^L$ must be financed by the floor remainders, which are relatively $O(1/x)$ in logarithms. The resulting bound on the cycle minimum excludes every period that is not admissible for the inequality, once a uniform logarithmic floor-error bound is available above a verified descent floor.

Throughout this section write $L = |w|$ and $o = \#O(w)$ for a cycle word w based at a cycle minimum $n \geq 2$. Natural logarithms are written $\log$. The unique one-step fibres of Section 3 give, for every state $x \geq 1$ with image $y = J(x)$,

$$y^2 \leq x^e < (y+1)^2, \quad e = \begin{cases} 1, & x \text{ even,} \\ 3, & x \text{ odd.} \end{cases}$$

Lemma 4.1 (dyadic-cell logarithm). If $z, y \geq 1$ and $z < (y+1)^2$, then $\log z \leq 2 \log y + 2/y$.

Proof. The hypothesis gives $\log z \leq 2 \log(y+1)$. The inequality $\log(1+u) \leq u$ for $u > 0$ yields $\log(y+1) = \log y + \log(1+1/y) \leq \log y + 1/y$. $\square$

Lemma 4.2 (one-step bounds). Let $x \geq 2$ and $y = J(x)$. If x is even, then $\log x \leq 2 \log y + 2/y$. If x is odd, then $3 \log x \leq 2 \log y + 2/y$.

Proof. The even cell is $y^2 \leq x < (y+1)^2$. The cell logarithm lemma on $z = x$ is the first claim. The odd cell is $y^2 \leq x^3 < (y+1)^2$. The same lemma on $z = x^3$ gives $\log(x^3) \leq 2 \log y + 2/y$. $\square$

On a cycle minimum every proper prefix is non-contracting: a prefix with $3^{\#O} < 2^k$ would satisfy $J^k(n) < n$ by Corollary 2.3, contradicting minimality. Thus $2^k \leq 3^{o_k}$ for every prefix of length k , where o_k is the odd count of that prefix.

Lemma 4.3 (unrolled envelope). Write $x_k = J^k(n)$ and o_k for the odd count of the length- k prefix. For every $0 \leq k \leq L$,

$$3^{o_k} \log n \leq 2^k \log x_k + \frac{k 3^{o_k}}{n}.$$

Proof. The case $k = 0$ is an equality. Suppose the claim holds at $k < L$, and write $x = x_k$ and $y = x_{k+1}$. Minimality gives $n \leq y$, and the prefix law gives $2^{k+1} \leq 3^{o_{k+1}}$, hence

$$\frac{2^{k+1}}{y} \leq \frac{3^{o_{k+1}}}{n}.$$

If the next letter is even, then $o_{k+1} = o_k$ and the one-step bound gives $\log x \leq 2 \log y + 2/y$. Multiply by 2^k and add the inductive remainder:

$$3^{o_k} \log n \leq 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k 3^{o_k}}{n} \leq 2^{k+1} \log y + \frac{(k+1) 3^{o_k}}{n}.$$

If the next letter is odd, then $o_{k+1} = o_k + 1$. Multiply the inductive bound by 3 and apply the one-step bound in the form $3 \log x \leq 2 \log y + 2/y$:

$$3^{o_{k+1}} \log n \leq 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k 3^{o_{k+1}}}{n} \leq 2^{k+1} \log y + \frac{(k+1) 3^{o_{k+1}}}{n}.$$

This is the claim at $k+1$. $\square$

Theorem 4.4 (finance). Let w be a cycle word of length L with o odd letters, based at a cycle minimum $n \geq 2$. Then

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

Proof. Apply Lemma 4.3 at $k = L$. Periodicity gives $x_L = n$ and $o_L = o$, so

$$3^o \log n \leq 2^L \log n + \frac{L \cdot 3^o}{n}.$$

The word is formally expanding, so $3^o > 2^L$. Rearranging and multiplying by n is the claim. $\square$

This is the conceptual centre of the note. Exact cells give a one-step logarithmic defect; cycle minimality lets the defect be unrolled against the cycle minimum; the formal surplus $3^o - 2^L$ must then be paid by a finite accumulated budget. Ideal dynamics expands and exact dynamics returns, so the floor errors finance the expansion. The inequality $\log(1 + u) \leq u$ is the only analytic input; the content is that interaction. The Lean form is exactly Theorem 4.4 (constant 1): `cycleMin_finance`.

Hierarchy of forms. These three layers must not be conflated.

1. *Theorem 4.4* is the conceptual sharp inequality. Constant 1:

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o,$$

equivalently $\theta \leq L/(n \log n)$ with $\theta = 1 - 2^L/3^o$. It charges every cell defect at the cycle minimum. Lean: `cycleMin_finance`.

2. *Corollary 4.4c* (below) is the same cell-log unroll with remainders kept as $1/x_{i+1}$. It is the strongest proved form of those defects. Lean: `cycleMin_finance_inv_sum`.
3. *Corollary 4.5* is the convenient length-only statewise bound: a verified descent floor N_0 plus a parity charge of the defect sum produces a per-length threshold $n_{\max}(L)$ and excludes every L with $n_{\max}(L) \leq N_0$.
4. *Theorem 4.6* is the numerical certification of Corollary 4.5 at $N_0 = 10^6$. The certified identity is the conservative relative-defect form

$$1 - \frac{2^L}{3^o} \leq \frac{6}{5} \sum_{i=1}^L \frac{1}{x_i \log x_i}.$$

The factor $6/5$ is a convenient uniform majorant of $-\log(1 - \delta)/\delta$ on $[0, 1/6]$. It is not Theorem 4.4, and the headline cutoff 25781 is not an artifact of that majorant.

The relative-defect unroll is a parallel identity, not Theorem 4.4 multiplied by $6/5$. Charging every state at the cycle minimum in that identity recovers the coarser comparison $\theta \leq (6/5)L/(n \log n)$, which is Theorem 4.4 with coefficient $6/5$. At this floor that uniform charge excludes only through length 1053. The computational table uses a stricter length-only parity charge of the same identity.

Corollary 4.4c (inv-sum). Let w be a cycle word of length L with o odd letters, based at a cycle minimum $n \geq 2$. Write $x_i = J^i(n)$. Then

$$(3^o - 2^L) \log n \leq 3^o \sum_{i=1}^L \frac{1}{x_i},$$

equivalently $\theta \leq (\sum_i 1/x_i)/\log n$.

Proof. The same induction as Lemma 4.3, keeping each cell defect as $2^{k+1}/x_{k+1}$ instead of replacing it by $3^{o_{k+1}}/n$. At $k = L$ one has $x_L = n$. Lean: `cycleMin_log_envelope_inv`, `cycleMin_finance_inv_sum`. $\square$

The computational table of Theorem 4.6 uses a weaker per-step bound, valid on every cycle because every start below 12 reaches 1 (so every cycle state is at least 12). For a state x with image $y = J(x) \geq 12$, the relative defect $\delta = (x^e - y^2)/x^e$ satisfies $\delta \leq 2/y \leq 1/6$. Writing $\varepsilon = -\frac{1}{2} \log(1 - \delta)$ and using $-\log(1 - \delta) \leq \frac{6}{5} \delta$

on $[0, 1/6]$ gives $\varepsilon \leq (6/5)/y$. Unrolling $t_{i+1} = (e_i/2)t_i - \varepsilon_i$ around the cycle then yields the conservative identity displayed in item 4 of the hierarchy.

On a minimum-based cycle of length L with o odd letters and $e = L - o$ even letters, the last letter is even, so $e \geq 1$ and the number of odd-run starts is at most e . Every even state satisfies $x \geq n^2$. Every odd state preceded by an odd state satisfies $x \geq t = \lfloor n^{3/2} \rfloor$. Therefore

$$\sum_{i=1}^L \frac{1}{x_i \log x_i} \leq \frac{e}{n \log n} + \frac{o-e}{t \log t} + \frac{e}{2n^2 \log n},$$

and

$$n \log n \left(1 - \frac{2^L}{3^o}\right) \leq \frac{6}{5} \left(e + (o-e) \frac{n \log n}{t \log t} + \frac{e}{2n}\right).$$

The optimal uniform coefficient on $[0, 1/6]$ is $6 \log(6/5) \approx 1.093$. Replacing $6/5$ by that constant, or even by 1, on the same parity charge does not change the first surviving length: one still has $n_{\max}(25780) \leq 10^6 < n_{\max}(25781)$. The published table and the count 141 keep the proved coefficient $6/5$. The cutoff 25781 is therefore not an artifact of an avoidable loss in the majorant.

Lemma 4.4b (odd-count monotonicity). Write $\theta(o) = 1 - 2^L/3^o$ and

$$R(o) = e + (o-e)\alpha + \frac{e}{2n}, \quad e = L - o, \quad \alpha = \frac{n \log n}{t \log t}.$$

The certified parity comparison is $n \log n \cdot \theta(o) \leq (6/5)R(o)$. For every $n \geq 12$ one has $\alpha < 1/2$, hence

$$R(o+1) - R(o) = 2\alpha - 1 - \frac{1}{2n} < 0.$$

Also $\theta(o+1) > \theta(o)$. Therefore if the comparison fails at some admissible o , it fails at every larger odd count. Equivalently, the largest n at which the comparison can hold occurs at the least admissible odd count $o_{\min}(L) = \min\{o : 3^o > 2^L\}$. The same monotonicity holds for any positive coefficient in place of $6/5$, and for the constant-1 comparison of Theorem 4.4.

Proof. The difference $R(o+1) - R(o)$ is the displayed coefficient of o . For $n \geq 12$ one has $t = \lfloor n^{3/2} \rfloor \geq n^{3/2} - 1 > 2n$, hence $\alpha < (n \log n)/(2n \log(2n)) = (\log n)/(2 \log(2n)) < 1/2$. The map $o \mapsto \theta(o)$ is strictly increasing on integers o with $3^o > 2^L$. If $n \log n \cdot \theta(o) > (6/5)R(o)$, then $n \log n \cdot \theta(o+1) > (6/5)R(o) > (6/5)R(o+1)$. $\square$

Define

$$\gamma(L) = \frac{3^{o_{\min}(L)}}{2^L} - 1,$$

and write $n_{\max}(L)$ for the largest integer n at which the displayed parity inequality can still hold at $o = o_{\min}(L)$. The coarser comparison $n \log n \leq (6/5)L \cdot 3^{o_{\min}}/(3^{o_{\min}} - 2^L)$ is used only as a check; Theorem 4.6 uses $n_{\max}(L)$ from the parity form. Lemma 4.4b is why that table may be computed at $o_{\min}$ alone.

Proposition 4.4a (finance-survivor algorithm). Fix a verified descent floor N_0 . For each integer $1 \leq L \leq 10^5$, compute $o_{\min}(L) = \min\{o : 3^o > 2^L\}$ by exact integer arithmetic and $n_{\max}(L)$ from the parity inequality, and retain L if and only if $n_{\max}(L) > N_0$. The resulting finance-survivor set is

$$\mathcal{E}(N_0) = \{L : 1 \leq L \leq 10^5, n_{\max}(L) > N_0\}.$$

The printed instance is $\mathcal{E} = \mathcal{E}(10^6)$, with $|\mathcal{E}| = 141$.

Corollary 4.5. If every integer $2 \leq n \leq N_0$ reaches 1, then no nontrivial cycle of length L exists whenever $n_{\max}(L) \leq N_0$.

Proof. A periodic state never reaches 1, so every cycle state is at least $N_0 + 1$. The length-only parity charge of the certified relative-defect identity then forces the minimum to satisfy the displayed inequality,

hence $n \leq n_{\max}(L)$. This is the convenient statewise bound in the hierarchy after Theorem 4.4; it is not a replacement for that theorem. $\square$

Record values of the parity $n_{\max}$ include $n_{\max}(19) = 133$, $n_{\max}(84) = 2323$, $n_{\max}(569) = 23568$, $n_{\max}(1054) = 788014$, and $n_{\max}(25781) = 26254995$.

Theorem 4.6 (verified computation). Every integer $2 \leq n \leq 10^6$ reaches 1. Consequently:

- (A) there is no nontrivial Juggler cycle of length at most 25780;
- (B) if a nontrivial cycle has period $L \leq 10^5$, then $L \in \mathcal{E}$, where $|\mathcal{E}| = 141$.

The bound closes continuously through 25780; 25781 is the first integer for which this particular inequality does not exclude a cycle. The coefficient $6/5$ is the certification majorant of item 4 in the hierarchy after Theorem 4.4, not the source of the cutoff.

Proof. The descent floor is Proposition 1.3. The gap table computes $o_{\min}(L)$ and $n_{\max}(L)$ for every $1 \leq L \leq 10^5$ by Lemma 4.4b. Corollary 4.5 at $N_0 = 10^6$ excludes every L with $n_{\max}(L) \leq 10^6$. The contiguous excluded prefix is $L \leq 25780$. The complementary set in range is $\mathcal{E}$; checksums are Appendix B. $\square$

Run packing

Theorem 4.7 refines the length-only charge by separating the unique minimum, 00E-scale valleys, 0E-scale valleys, internal odds, and evens. It is a supporting comparison at the same floor; the leftover count $141 \rightarrow 99$ does not raise the cutoff. A cycle cannot put an 0E-start at the minimum (Theorem 3.2), and an n -circuit of k odds and ℓ evens stays at least n only if $3^k \geq 2^{k+\ell}$.

Theorem 4.7 (run-type packing). Let w be a cycle word of length L with $o = o_{\min}(L)$ odd letters and $e = L - o$ even letters, based at a cycle minimum $n \geq 12$. Write v for the least odd integer with $v^3 \geq n^4$, and write $t = \lfloor n^{3/2} \rfloor$. Then $o - e < e$, the largest number of n -scale valleys compatible with the even cap is $o - e$ copies of 00E, and the remaining $2e - o$ circuits are 0E from v . The cycle minimum occurs once, so

$$\sum_{i=1}^L \frac{1}{x_i \log x_i} \leq \frac{1}{n \log n} + \frac{o - e - 1}{(n + 2) \log(n + 2)} + \frac{2e - o}{v \log v} + \frac{1}{t \log t} + \frac{o - e - 1}{t_+ \log t_+} + \frac{e}{2n^2 \log n},$$

where $t_+ = J(n + 2)$. Combined with the $6/5$ unroll this is strictly smaller than the parity sum of Corollary 4.5 whenever $2e - o > 0$. Sending the cycle maximum to infinity removes one even term and does not change the valley packing.

Proof. The statement is at $o = o_{\min}(L)$. Lemma 4.4b already excludes every larger odd count from the parity comparison used in Theorem 4.6; the packing is a refinement at that same odd count. Formal expansion at $o_{\min}$ forces $3o < 2L$ on every leftover length in range, equivalently $o - e < e$. The even-cap comparison $3^k \geq 2^{k+\ell}$ is the ideal power envelope; floors only help. An 0E-start is followed by an even state, so Theorem 3.2 gives $J(v) \geq n^2$ and therefore $v^3 \geq n^4$. Unique visit of the cycle minimum is periodicity. The displayed sum charges one cheap valley at n , the remaining $o - e - 1$ cheap valleys at the next odd integer $n + 2$, the expensive valleys at v , one internal odd at t , the remaining internals at $J(n + 2)$, and every even at n^2 . Any deeper odd run or any higher valley only decreases the sum. $\square$

Theorem 4.8 (run-type table). At the verified descent floor $N_0 = 10^6$, the packed comparison of Theorem 4.7 excludes the 42 lengths $56347 + 1054k$ for $k = 0, \dots, 41$ and leaves an explicit set $\mathcal{E}_{\text{run}} = \mathcal{E}_{\text{run}}(10^6)$ of 99 lengths. The first survivor remains 25781. In particular, if a nontrivial cycle has period $L \leq 10^5$, then $L \in \mathcal{E}_{\text{run}}$.

Proof. The 141 lengths of Theorem 4.6(B) are tested at $n = 10^6 + 1$ against the packed right-hand side. The comparison is certified on each length: no entry is left uncertain. The 42 excluded lengths are exactly the arithmetic progression named in the statement. The complementary set in that range is $\mathcal{E}_{\text{run}}$. Checksums are Appendix B. The period cutoff remains 25781. $\square$

Arithmetic structure of the finance survivors

The leftover lengths cluster around the continued-fraction approximants of $\log 2 / \log 3$. That organizes the table; it does not constrain a hypothetical cycle.

Proposition 4.9 (finance-survivor lattice). Write $v_* = (25781, 16266)$ and $v_{1054} = (1054, 665)$. Then

$$25781 \cdot 665 - 1054 \cdot 16266 = 1,$$

so these two vectors are a unimodular basis of $\mathbb{Z}^2$. Every length in $\mathcal{E}_{\text{run}}$, and every one of the 42 packing deaths of Theorem 4.8, is of the form $(L, o_{\min}) = a v_* + b v_{1054}$. With the table cap $L \leq 10^5$ the 99 survivors fall in three affine slices of counts 29, 47, and 23. The identification with $\mathcal{E}_{\text{run}}$ is Theorem 4.8. This is a change of coordinates for (L, o) , not a relation between hypothetical cycles.

Proof. The determinant identity is integer arithmetic. The generator comparison $3^{665} > 2^{1054} > 3^{664}$ gives $o_{\min}(1054) = 665$. Direct evaluation of $o_{\min}$ on the 141 lengths of Theorem 4.6(B) places each pair $(L, o_{\min})$ on the displayed lattice, with the 42 packing deaths the F_1 continuation $b \geq 29$. $\square$

5. Limitations and future directions

The result does not imply termination. The remaining finance-survivor lengths are uncontrolled, and existence of a cycle at such a length is open.

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$. Even starts realize E ; an odd start with even image realizes OE . The complement of those two words is the odd-to-odd class. If every start above 1 has some descent certificate, strong induction yields arrival at 1. That hypothesis is not proved, and a uniform run bound on expanding blocks is unavailable: four consecutive expanding blocks occur already at

$$1999 \xrightarrow{OOE} 5169 \xrightarrow{OOOOEE} 50093 \xrightarrow{OOE} 193753 \xrightarrow{OOE} 887471.$$

The next concrete direction is the number p of odd runs on a minimum-based cycle. The run form already gives $p \leq e$ and, because the first odd run has length at least two, $p \leq o-1$, hence $p \leq \min(e, o-1) < 0.3691 L$ on an expanding word. That is only the trivial ceiling. A genuine lower bound on p , or a peak-height / peak-count tradeoff, would feed Theorem 4.7. Neither is proved here.

The same pattern — a piecewise power map, integer rounding, and a cycle minimum — produces a defect-financing obstruction. The Juggler-specific content is the interaction of $x^{3/2}$ and $x^{1/2}$. Analogous questions for other piecewise floor-power maps are not taken up here.

Lean names are in Appendix A. The computational certificate is Proposition 1.3.

Appendix A. Lean names

The core mathematical lemmas of Sections 2–4 are mechanized in Lean 4. The names below are the corresponding theorems in `formal/Problems/Juggler/`, imported by `Problems.JugglerPaper`. Selected finite classifications of Section 3 are `native_decide` tables (Appendix D). Theorems 4.6 and 4.8 are independently certified computations. Proposition 4.9’s arithmetic is Lean; its identification with $\mathcal{E}_{\text{run}}$ is Theorem 4.8. This is not a claim that the paper as a whole is formally verified.

Text	Lean
itinerary semantics	<code>follows_iff_word</code> , <code>image_eq_iterate</code> , <code>image_append</code>
Theorem 2.1	<code>image_monotone_of_follows</code>
Theorem 2.2	<code>power_bound_word</code>
Corollary 2.3	<code>power_bound_contracts</code>

Text	Lean
Theorem 2.4	global_defect_identity
Theorem 2.5	global_defect_eq_zero_iff_localsTight, global_defect_eq_zero_implies_monochrome, power_bound_eq_iff_extremal
Theorem 2.6	global_defect_append
Corollary 2.7	image_eq_start_defectRatio
per-step slack	one_plus_eta_lt_succ_sq
Lemma 3.1	odd_cell_unique
Theorem 3.2	cycle_word_formally_expanding, cycleMin_start_odd, cycleMax_start_even, cycleMin_not_end_odd, square_scale_superquadratic, cycleMin_to_even_superquadratic
Lemma 3.3	lower_growth_word
Lemma 3.4	oo_suffix_threshold, ooo_suffix_threshold, threshold_inherits_odd_append, cycle_last_even_interval, no_cycle_odd_run_append_even
Lemma 3.5	no_cycle_word_oooeoe, no_cycle_word_ooooee
Theorem 3.6	no_cycle_word_length_le_six
Lemma 3.7	no_cycle_word_ooooeoe, no_cycle_word_ooooeee
Theorem 3.8	no_cycle_word_length_le_seven, with no_cycle_word_ooeooooe, no_cycle_word_oooeoooo
Lemma 3.9	cycle_trailing_evens_lt
Lemma 3.10	lowerDenom_replicate_odd, odd_run_lower_growth
Lemma 3.11	no_follows_seven_odds_of_lt256
Theorem 3.12	no_cycle_word_two_even_ee, no_cycle_word_two_even_eoe
Theorem 3.13	no_cycleMin_gapped_three_even_ee, no_cycleMin_gapped_three_even_eoe
Theorem 3.14	no_cycle_word_three_even_eee, with no_cycle_word_oooooooeee
Theorem 3.15	no_cycle_word_three_even_eoeoe
Theorem 3.16	no_cycle_word_three_even_eooeee
Theorem 3.17	no_cycle_word_three_even_eooooee
Theorem 3.18	no_cycle_word_three_even_eoeoe
Theorem 3.19	no_cycle_word_three_even_eoeoe
Theorem 3.20	no_cycle_word_three_even_eooeoe
Theorem 3.21	no_cycle_word_gapped_three_even_ee, no_cycle_word_gapped_three_even_eoe
Theorem 3.22	no_cycle_word_even_count_le_three
Corollary 3.23	cycle_word_length_ge_eleven
Lemma 3.21b	canonical run form; Theorem 3.2
Lemma 3.21a	the case split of Theorem 3.22
Lemma 4.1	log_le_two_log_add
Lemma 4.2	log_step_even, log_step_odd
Lemma 4.3	cycleMin_log_envelope
Theorem 4.4	cycleMin_finance
Corollary 4.4c	cycleMin_log_envelope_inv, cycleMin_finance_inv_sum
Lemma 4.4b	odd-count monotonicity; human proof, not Lean

Text	Lean
Theorem 4.7	run-type packing; human proof, not Lean
Theorem 4.8	run-type table; verified computation, not Lean
Proposition 4.9	run_survivor_unimodular, run_survivor_seed_F2, run_survivor_seed_F3, three_pow_step_gt_two_pow_step, runSurvivors_length
short certificates (Section 5)	even_finiteProgress, odd_even_finiteProgress
no certificate $\Rightarrow$ odd-to-odd	no_finiteProgress_implies_odd_odd
induction to 1	reachesOne_of_all_finiteProgress
four-block chain	four_block_pe_1999

Appendix B. Admissible lengths

The record lengths of $\gamma(L)$ through 10^5 , with the parity 6/5 bound $n_{\max}$ of Section 4, are

L	$o_{\min}$	$n_{\max}$
1	1	3
3	2	7
11	7	25
19	12	133
84	53	2323
569	359	23568
1054	665	788014
25781	16266	26254995
50508	31867	162848325

At the verified descent floor $N_0 = 10^6$ the first seven rows are excluded. The set $\mathcal{E} = \mathcal{E}(10^6)$ is defined by Proposition 4.4a: for each $1 \leq L \leq 10^5$, compute $o_{\min}(L)$ by exact integer arithmetic and $n_{\max}(L)$ from the parity inequality, and retain L if and only if $n_{\max}(L) > 10^6$. This produces 141 lengths. The first few are

$$25781, 26835, 27889, 28943, 29997,$$

and a typical combination is $26835 = 25781 + 1054$. The last is 99561. The complete list is the `lengths` array of `data/research/juggler/cycle_finance/exceptions_parity.json`. The SHA-256 of that array, serialized as a JSON list of integers with no spaces, is `dd71aa1527656ba51cb031bafa5497f7bfdbbc43151ffba2c595793326bf7944`. The SHA-256 of the whole file `exceptions_parity.json` is `6b4eec79295b70cdeb9f7db677b7fd57bcb9bd1b51177e1e74aad7b`. The SHA-256 of the first-passage file `floor.json` in the same directory is `5b1ce1eec61301cf5b4f969cd5b58954255194e5c7b2`. The parity table is written by `research.juggler_sequence.cycle_finance.write_parity_artifacts`. The first-passage file is regenerated by `python -m research.juggler_sequence.cycle_finance`.

The run-type table of Theorem 4.8 is `data/research/juggler/cycle_finance/budget_opt.json`. The 42 excluded lengths are `killed_by_budget`; their SHA-256, serialized as a JSON list of integers with no spaces, is `9d108776d6dc5dc1ae2594058850463cd2d3995cc57b9811471f08e3b818b90a`. The complementary 99 lengths are $\mathcal{E}_{\text{run}}$. Their SHA-256, in the same serialization, is `9e2098923ccb39933630b116133a3fc2ddaf98ace4eb76dbab9b5`. The first few survivors remain

$$25781, 26835, 27889, 28943, 29997;$$

the last survivor is 99477. The three families and the unimodular basis are Proposition 4.9. The finite leftover tables of Section 3 are the Lean `native_decide` evaluations named in Appendix A (`LeftoverEval.lean`, `LeftoverShort.lean`, `LeftoverFamilies.lean`). The lattice arithmetic of Proposition 4.9 is `RunSurvivorLattice.lean`.

Appendix C. Exact floor defect

This appendix records the exact identity behind Theorem 2.2. The present note uses only the nonnegativity $\Delta_w(n) \geq 0$. A nontrivial itinerary-dependent lower bound on Δ_w is left for future work.

For a single branch,

$$x^e = J(x)^2 + \rho(x), \quad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \leq \rho(x) < 2J(x) + 1$. Write $\text{gap}(a, \rho, e) = (a + \rho)^e - a^e$. The *global defect* $\Delta_w(n)$ is the terminal value of the resulting power-gap recurrence.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \quad \Delta_w(n) \geq 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \geq 0$.

Proof. Induct on w . The empty word is $n = n + 0$. An even letter substitutes $x = J(x)^2 + \rho(x)$ into the inductive identity and lifts the new remainder through 2^ℓ . An odd letter cubes the identity and then substitutes $x^3 = J(x)^2 + \rho(x)$. Each step adds a nonnegative power-gap. $\square$

For a one-letter illustration, $n = 3$ and $w = O$ give $J(3) = 5$ and $3^3 = 27 = 5^2 + 2$, so $\Delta_O(3) = 2$.

Theorem 2.5 (vanishing). If w is realized at n , the following are equivalent: $\Delta_w(n) = 0$; every local remainder along w vanishes; and $(J^{|w|}(n))^{2^{|w|}} = n^{3^{\#O(w)}}$. In that case w is monochrome: either $w = E^k$ and $n = a^{2^k}$ for an even a , or $w = O^k$ and $n = a^{2^k}$ for an odd a . A realized mixed word therefore has $\Delta_w(n) > 0$.

Proof. Theorem 2.4 gives the first and third items. A power-gap vanishes if and only if its addend vanishes, so zero defect forces every remainder to vanish, and conversely. Vanishing remainders preserve parity, hence monochrome itineraries, and unique factorization produces the two power towers. $\square$

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then $\Delta_{uv}(n)$ is the sum of two power-gaps, one lifting $\Delta_u(n)$ through $3^{\#O(v)}$ and one lifting $\Delta_v(m)$ through $2^{|u|}$. In particular $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$, and every local remainder satisfies $\rho_j^{2^j} \leq \Delta_w(n)$.

Proof. Apply Theorem 2.4 to u , to v , and to uv , and expand the two power-gaps. $\square$

Composition is polynomial, not additive. The Lean form is `global_defect_append`.

Corollary 2.7 (cycle surplus). If w is a cycle word at n , then $\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$ exactly.

Proof. Theorem 2.4 with $m = n$. $\square$

A mixed word has strict total defect, but the relative slack of a single letter tends to 0 with the state, so no uniform local tax exists. Theorem 4.4 does not use these identities.

Appendix D. Family exclusions

This appendix records the family-by-family exhaustion used by Lemma 3.21a and Theorem 3.22. Each geometry is killed by a next-square obstruction, by long odd-run growth against a last-even cell, or by a finite exceptional window.

Theorem 3.12 (two-even leftover families). Let $k \geq 6$ and $n \geq 2$. Neither $O^{k-2}EE$ nor $O^{k-3}EOE$ is a cycle word at n .

Proof. * First, if $n \geq 256$, then

$$n^{3^{k-2}} > 2^{e_{k-2}}(n+1)^{2^k}.$$

The case $k = 6$ is the tail inequality of Lemma 3.5. If the display holds at some $k \geq 6$, cubing both sides produces

$$n^{3^{k-1}} > 2^{3e_{k-2}}(n+1)^{3 \cdot 2^k}.$$

The recurrence of Lemma 3.10 gives $e_{k-1} = 3e_{k-2} + 2^{k-1}$, so the desired comparison at length $k+1$ reduces to $2^{2^{k-1}} < (n+1)^{2^k}$. Equivalently $2 < (n+1)^2$, which holds for every $n \geq 256$.

Now suppose $O^{k-2}EE$ is a cycle word at such an n . Write $z = J^{k-2}(n)$. Lemma 3.9 with $r = 2$ gives $z < (n+1)^4$. Lemma 3.10 on the prefix O^{k-2} gives $n^{3^{k-2}} \leq 2^{e_{k-2}} z^{2^{k-2}}$, hence $n^{3^{k-2}} < 2^{e_{k-2}} (n+1)^{2^k}$, contradicting the tail.

Finally suppose $O^{k-3}EOE$ is a cycle word at such an n . Write $z = J^{k-3}(n)$ and $y = \lfloor \sqrt{z} \rfloor$, so $z < (y+1)^2$. Lemma 3.10 on O^{k-3} and cubing produce $n^{3^{k-2}} < 2^{3e_{k-3}} (y+1)^{3 \cdot 2^{k-2}}$. The last letters OE give the odd-cell bound $y^3 < (n+1)^4$. The comparison $(y+1)^3 < 2(n+1)^4$ of Lemma 3.5 applies at this scale. Raising it to the power 2^{k-2} and using $e_{k-2} = 3e_{k-3} + 2^{k-2}$ recovers again $n^{3^{k-2}} < 2^{e_{k-2}} (n+1)^{2^k}$.

For $2 \leq n < 256$, the cases $k = 6$ and $k = 7$ are Lemmas 3.5 and 3.7. The remaining short words O^6EE , O^5EOE , and O^6EOE fail to return on the same 254-start window; this is the Lean `native_decide` evaluation behind `no_cycle_word_two_even_ee` and `no_cycle_word_two_even_eoe` (Appendix A). Any longer leftover of either family begins with seven consecutive odd letters, which Lemma 3.11 forbids on this window. $\square$

Theorem 3.13 (first-even transport). Let $n \geq 2$. No minimum-based cycle word at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. * Write $y = J^{a+1}(n)$ for the state after the first even letter. Minimum-basedness gives $y \geq n$. In the first family the remainder after that letter is O^bEE with $b+2 \geq 6$; in the second it is O^bEOE with $b+3 \geq 6$. The trailing-even and last-odd cells of those remainders are measured against the cycle start n . Combined with Lemma 3.10 at the remainder start y , the same algebra as in Theorem 3.12 produces

$$y^{3^{\ell-2}} < 2^{e_{\ell-2}} (n+1)^{2^\ell} \leq 2^{e_{\ell-2}} (y+1)^{2^\ell},$$

where ℓ is the remainder length. If $y \geq 256$, the first paragraph of Theorem 3.12 supplies the opposite inequality at y .

If $y < 256$, then $n \leq y < 256$. A gapped leftover of total length at least 17 has $a \geq 7$ or $b \geq 7$, so either the prefix or the remainder realizes seven consecutive odd letters, contradicting Lemma 3.11. The finitely many short-gap words with $2 \leq a \leq 6$ and $b \leq 6$ fail to be minimum-based cycle words on the window $2 \leq n < 256$; this is the Lean `native_decide` evaluation behind `no_cycleMin_gapped_three_even_ee` and `no_cycleMin_gapped_three_even_eoe` (Appendix A). $\square$

Theorem 3.14 (three trailing evens). Let $a \geq 6$ and $n \geq 2$. The word O^aEEE is not a cycle word at n .

Proof. * Write $z = J^a(n)$. Lemma 3.9 with $r = 3$ gives $z < (n+1)^8$. Lemma 3.10 then yields $n^{3^a} < 2^{e_a} (n+1)^{2^{a+3}}$ on any such cycle word. For $n \geq 128$ the opposite comparison

$$n^{3^a} > 2^{e_a} (n+1)^{2^{a+3}}$$

holds. The case $a = 6$ is $n^{729} > 2^{1330} (n+1)^{512}$. Indeed, for $n \geq 128$ one has $(n+1)^{512} < (129/128)^{512} n^{512} < e^4 n^{512} < 64 n^{512}$, so the claimed bound reduces to $n^{217} > 2^{1336}$. Since $n \geq 128 = 2^7$, the left side is at least 2^{1519} . Cubing the $a = 6$ comparison produces the general case: the recurrence of Lemma 3.10 reduces the inductive step to $(n+1)^4 > 2$, which holds for every $n \geq 128$.

For $2 \leq n < 128$, the case $a = 6$ is a table of evaluations of `OOOOOOEEE`: at every such start the word fails to return. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooooooee` (Appendix A). For $a \geq 7$ the prefix contains seven consecutive odd letters, which Lemma 3.11 forbids on this window. $\square$

Theorem 3.15 (mixed bunched family $EOEE$). Let $a \geq 5$ and $n \geq 2$. The word O^aEOEE is not a cycle word at n .

Proof. * First let $n \geq 4$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and $p = J(y)$. Lemma 3.9 with $r = 2$ after the prefix O^aEO gives $p < (n+1)^4$. The letter after y is odd, so Lemma 3.10 at length one yields $y^3 \leq 4p^2 < 4(n+1)^8$.

For $n \geq 4$ one has $4(n+1)^8 < (n+1)^9$, hence $y < (n+1)^3$. The even cell at z then gives $z < (y+1)^2 \leq (n+1)^6$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{6 \cdot 2^a}.$$

For $a = 5$ and $n \geq 314$, the opposite comparison $n^{243} > 2^{422}(n+1)^{192}$ holds. The base instance is the finite inequality $2^{422}315^{192} < 314^{243}$. If the display holds at some $n \geq 1$, the elementary comparison $n(n+2) < (n+1)^2$ upgrades it to the same display at $n+1$. Cubing then produces the comparison at $a+1$, once $(n+1)^6 > 4$. In particular the case $a = 6$ already holds for every $n \geq 16$.

For $2 \leq n < 314$ and $a = 5$, and for $2 \leq n < 16$ and $a = 6$, the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eooo` (Appendix A). For $a \geq 7$ and $n < 256$, Lemma 3.11 applies. For $a \geq 6$ and $n \geq 16$, the tail of the previous paragraph applies. $\square$

Theorem 3.16 (mixed bunched family $EOOEE$). Let $a \geq 4$ and $n \geq 2$. The word O^aEOOEE is not a cycle word at n .

Proof. * First let $n \geq 32$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and p for the image after the prefix O^aEOO . Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The two letters after y are odd, so Lemma 3.10 at length two yields $y^9 \leq 2^{10}p^4 < 2^{10}(n+1)^{16}$. For $n \geq 32$ one has $2^{10} < (n+1)^2$, hence $y < (n+1)^2$. The even cell at z then gives $z < (y+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a + 2$.

For $2 \leq n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoooo` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.17 (mixed bunched family $EOOOEE$). Let $a \geq 3$ and $n \geq 2$. The word $O^aEOOOEE$ is not a cycle word at n .

Proof. * First let $a \geq 4$ and $n \geq 3$, and write $z = J^a(n)$, $y = \lfloor \sqrt{z} \rfloor$, and p for the image after the prefix O^aEOOO . Lemma 3.9 with $r = 2$ gives $p < (n+1)^4$. The three letters after y are odd, so Lemma 3.10 at length three yields $y^{27} \leq 2^{38}p^8 < 2^{38}(n+1)^{32}$. For $n \geq 3$ one has $2^{38} < (n+1)^{22}$, hence $y < (n+1)^2$. The even cell at z then gives $z < (y+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the opposite of the shared tail of Theorem 3.12 at length $k = a + 2$, already used in Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The same two-even cell gives $z < (y+1)^2$, so Lemma 3.10 at the prefix O^3 yields $n^{27} < 2^{38}(y+1)^{16}$. The three-odd envelope on y still gives $y^{27} < 2^{38}(n+1)^{32}$.

If $y < 39$, then $n^{27} < 2^{38} \cdot 39^{16}$. The numerical comparison $2^{38} \cdot 39^{16} < 24^{27}$ contradicts $n \geq 24$.

If $y \geq 39$, the numerical comparison $40^{27} < 2 \cdot 39^{27}$ upgrades to $(y+1)^{27} < 2y^{27}$, hence $(y+1)^{27} < 2^{39}(n+1)^{32}$. Cubing the prefix bound three times produces $n^{729} < 2^{1026}(y+1)^{432}$. Raising the successor bound to the sixteenth power produces $(y+1)^{432} < 2^{624}(n+1)^{512}$. Combining these displays yields $n^{729} < 2^{1650}(n+1)^{512}$. The opposite comparison holds at $n = 197$ and persists to every larger start by the elementary comparison $n(n+2) < (n+1)^2$ already used in Theorem 3.15.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooooo` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.18 (mixed bunched family $EEOE$). Let $a \geq 5$ and $n \geq 2$. The word O^aEEOE is not a cycle word at n .

Proof. * First let $n \geq 4$, and write $z = J^a(n)$ and y for the last odd letter of O^aEEOE . The suffix $EEOE$ is a cycle suffix, so the last-odd cube of Theorem 3.15 gives $y^3 < (n+1)^4$. The two letters between z and y are

even, so $z < (y+1)^4$. For $n \geq 4$ the successor comparison $(y+1)^3 < 2(n+1)^4$ upgrades this to $z < (n+1)^6$. Combined with Lemma 3.10, any such cycle word would satisfy the same display as Theorem 3.15:

$$n^{3^a} < 2^{e_a}(n+1)^{6 \cdot 2^a}.$$

The opposite comparison is therefore the tail of Theorem 3.15: it holds for $a = 5$ and $n \geq 314$, and already for $a = 6$ and $n \geq 16$.

For $2 \leq n < 314$ and $a = 5$, and for $2 \leq n < 16$ and $a = 6$, the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eeoe` (Appendix A). For $a \geq 7$ and $n < 256$, Lemma 3.11 applies. For $a \geq 6$ and $n \geq 16$, the tail of the previous paragraph applies. $\square$

Theorem 3.19 (mixed bunched family $EOEOE$). Let $a \geq 4$ and $n \geq 2$. The word O^aEOEOE is not a cycle word at n .

Proof. * First let $n \geq 32$, and write $z = J^a(n)$, $w = \lfloor \sqrt{z} \rfloor$, and y for the last odd letter. The suffix EOE again gives $y^3 < (n+1)^4$. The one-odd envelope on w yields $w^3 \leq 4s^2$, where s is the image after O^aEO . The last-odd cell and $n \geq 32$ upgrade this to $w < (n+1)^2$, hence $z < (w+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For $n \geq 256$ and $a \geq 4$, this is the shared tail already used in Theorem 3.16.

For $2 \leq n < 256$, the cases $a = 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.20 (mixed bunched family $EOOEOE$). Let $a \geq 3$ and $n \geq 2$. The word $O^aEOOEOE$ is not a cycle word at n .

Proof. * First let $a \geq 4$ and $n \geq 4$, and write $z = J^a(n)$, $u = \lfloor \sqrt{z} \rfloor$, and y for the last odd letter. The suffix EOE gives $y^3 < (n+1)^4$, hence $(y+1)^3 < 2(n+1)^4$. The two letters after u are odd, so Lemma 3.10 at length two yields $u^9 \leq 2^{10}s^4 < 2^{10}(y+1)^8$. Cubing that display against the last-odd successor bound produces $u^{27} < 2^{38}(n+1)^{32}$. The same comparison as in Theorem 3.17 then gives $u < (n+1)^2$, hence $z < (u+1)^2 \leq (n+1)^4$. Combined with Lemma 3.10, any such cycle word would satisfy the shared two-even tail of Theorem 3.16.

Now let $a = 3$ and $n \geq 256$. The prefix O^3 against $z < (u+1)^2$ yields $n^{27} < 2^{38}(u+1)^{16}$, and the two-odd plus last-odd geometry of the previous paragraph yields $u^{27} < 2^{38}(n+1)^{32}$. These are the same two displays as in the $a = 3$ case of Theorem 3.17, with u in place of y , and the same small/large split applies.

For $2 \leq n < 256$, the cases $a = 3, 4, 5, 6$ fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eeoe` (Appendix A). For $a \geq 7$, Lemma 3.11 applies. $\square$

Theorem 3.21 (gapped leftovers as cycle words). Let $n \geq 2$. No cycle word at n has the form O^aEO^bEE with $a \geq 2$ and $b \geq 4$, or the form O^aEO^bEOE with $a \geq 2$ and $b \geq 3$.

Proof. * Every cycle word has a minimum-based rotation. It is therefore enough to check that every cyclic shift of either word is an already-excluded cycle-minimum orientation.

Write w for the gapped word. In the first family, $|w| = a + b + 3$. The rotation by $k = 0$ is the original word, excluded as a cycle minimum by Theorem 3.13. The rotation by $k = a + 1$ is the bootstrap word O^bEEO^aE . That word has an internal even letter and last gap at least 2. If $a \geq 3$, the last-gap threshold of Lemma 3.4 at OOO and $N = 3$ excludes it. If $a = 2$, the same lemma at OO and $N = 5$ excludes every start $n \geq 5$; the remaining odd start $n = 3$ does not realize four consecutive odd letters, so it cannot follow O^b for $b \geq 4$. The rotation by $k = a + b + 2$ begins with the last even letter of w , which Theorem 3.2 forbids at a cycle minimum. Every other rotation ends with an odd letter, likewise forbidden at a cycle minimum.

In the second family, $|w| = a + b + 4$. The same four classes appear: the original word is Theorem 3.13; the rotation by $k = a + 1$ is O^bEOEO^aE , excluded by the same last-gap thresholds, with the remaining start $n = 3$ and $a = 2$ either failing to realize four odds or, when $b = 3$, reaching 6 after $OOOE$ and then meeting

an odd letter; the rotation by $k = a + b + 2$ begins OE , forbidden by Theorem 3.2; and every other rotation ends odd.

The original start need not be a cycle minimum. After rotation the start is a minimum, so the hypothesis $y < n$ that blocked Theorem 3.13 does not arise. $\square$

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